

Ahmad Salehiyan

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Stillwater, OK | Oklahoma State University | [Google Scholar](#) | [LinkedIn](#)

Research Profile

Ph.D. researcher in Industrial Engineering and Management at Oklahoma State University (expected Summer 2027), with concurrent graduate training in Applied Statistics. Research combines machine learning, deep reinforcement learning, and stochastic optimization for data-driven decision-making under uncertainty. Published work includes an analysis of comorbidity and mortality patterns in the MIMIC-IV electronic health records dataset (179,524 adult patients) and transformer-based deep learning models in two peer-reviewed journals. Seeking to apply predictive modeling and decision-support methods to health data analytics.

Research interests: health data analytics and electronic health records; predictive modeling and risk stratification; deep learning and deep reinforcement learning; sequential decision-making under uncertainty (MDP/POMDP); stochastic optimization.

Education

- **Oklahoma State University**

Ph.D., Industrial Engineering and Management

Expected Summer 2027

- **Advisor:** Dr. Akash Deep (School of Industrial Engineering and Management)
- **Dissertation (proposed):** “Sequential Decision-Making Under Uncertainty in Smart Engineering Systems.”
- **Research focus:** decision-making from multiple noisy data streams under partial observability (POMDP), deep reinforcement learning, and multi-agent planning.

- **Oklahoma State University**

M.S., Applied Statistics

Expected 2026

- **Focus:** experimental design, statistical programming (R/SAS), data analytics, and statistical inference.
- **Relevant coursework:** Design of Experiments, Mathematical Statistics I & II, Statistical Programming (R/SAS), Applied Analytics.

- **K. N. Toosi University of Technology, Tehran**

M.S., Industrial Engineering

2019–2022

- **Thesis:** “Predictive Maintenance of Advanced Industrial Machines Using AI Techniques” (advisor: Dr. Abdollah Aghaie).

- **Islamic Azad University, Qazvin**

B.S., Industrial Engineering

2014–2019

- Ranked 4th in cohort (GPA: 3.75/4.0).

Technical Skills

- **Analytics programming:** Python (NumPy, SciPy, pandas, PyTorch, Gym), R, SAS, Julia, GAMS, Pyomo.
- **Databases and large-scale data:** SQL and PostgreSQL (relational databases); Apache Spark / PySpark (coursework level); high-performance computing cluster; GPU model training (dual NVIDIA RTX 6000 Ada workstation).
- **Machine learning and deep learning:** Transformer architectures (the model family underlying large language models), generative adversarial networks, autoencoders, deep Q-networks (DQN, DDQN), k-means clustering, association rule mining (Apriori), t-SNE, anomaly detection, time-series feature extraction.
- **Health data:** electronic health records (MIMIC-IV), coded diagnosis data, comorbidity and mortality analysis.
- **Sequential decision-making and optimization:** MDP, POMDP, value and policy iteration, Monte Carlo simulation, mixed-integer and linear programming, Benders and Lagrangian decomposition.
- **Research tools:** Git/GitHub, $\LaTeX$, simulation environments, experiment logging.

Research Experience

- **Graduate Research Assistant, Oklahoma State University** Aug 2023 – Present
 - Co-authored an association-rule-mining and clustering study of comorbidity and mortality patterns in the MIMIC-IV electronic health records dataset (179,524 adult patients, 846 distinct diseases); published in the IISE Annual Conference proceedings (2025).
 - Designed and implemented a **POMDP-based framework** for condition-based maintenance using multiple sensor streams; derived control-limit policies for operate-versus-preventive-maintenance decisions.
 - Built a hierarchical **Double Deep Q-Network (DDQN)** framework for warehouse picker routing and ran its training pipeline on a dual-GPU (NVIDIA RTX 6000 Ada) workstation.
 - Built discrete-event simulation to evaluate long-run average cost, Value-at-Risk (VaR), and Conditional Value-at-Risk (CVaR) under different policies; performed sensitivity analysis on sensor noise, policy thresholds, and cost parameters.
 - Formulated and solved mixed-integer programming models for multi-unit preventive-maintenance scheduling under technician and time-window constraints.
- **Research Assistant, K. N. Toosi University of Technology** Oct 2018 – Apr 2020
 - Engineered time-domain, frequency-domain, and wavelet-based features from high-dimensional sensor signals for early fault detection in industrial machinery.
 - Used feature-importance analysis to identify critical signals and designed statistical thresholds for fault diagnosis.

Selected Research Projects

- **Disease Cluster Analysis in Electronic Health Records (MIMIC-IV)**
 - Analyzed a processed cohort of 179,524 unique adult patients (ages 18–91) with 11,733 mortality cases and 846 distinct diseases, using patient, diagnosis, age, gender, and mortality fields.
 - Applied the Apriori algorithm (minimum support 0.01) to mine disease co-occurrence rules, evaluated by support, confidence, and lift; identified 628 unique association rules with confidence above 50%.
 - Grouped comorbidity rules with k-means clustering after t-SNE dimensionality reduction; selected $k = 4$ by silhouette analysis (silhouette score 0.44) and profiled clusters by mortality rate, age, and gender.
 - Computation performed in part at the OSU High-Performance Computing Center. Published in the IISE Annual Conference & Expo proceedings (2025); presented as a talk.
- **Transformer-Based Deep Learning for Anomaly and Intrusion Detection**
 - Developed a hybrid deep learning framework combining a Transformer, a generative adversarial network (GAN), and an autoencoder for intrusion detection in edge and industrial IoT systems; evaluated on three public benchmark datasets (WUSTL-IIoT-2021, EdgeIIoTset, TON_IoT). Published in *Future Internet* (2025).
 - Developed an evolutionary-optimized Transformer and deep reinforcement learning framework for false data injection detection in smart water infrastructure. Published in *Computers, Materials & Continua* (2026).
- **Multi-Sensor Condition-Based Maintenance under Partial Observability (POMDP)**
 - Modeled equipment health evolution under partial sensor observations as a Partially Observable Markov Decision Process and derived control-limit policies.
 - Implemented belief-state aggregation and state-space compression to scale to high-dimensional sensor data.
 - Validated on synthetic data and real condition-monitoring signals; showed 10–18% cost reduction versus calendar-based maintenance.
 - Status: preprint on TechRxiv (2024); manuscript under review at *INFORMS Journal on Data Science*.
- **Hierarchical Deep Reinforcement Learning for Warehouse Picker Routing**
 - Decomposed the global routing problem into aisle-level subproblems solved by dynamic programming, with a DDQN agent sequencing aisles.
 - Benchmarked against standard heuristics (S-shape, largest-gap); achieved 3.0%–15.4% improvement in total travel distance.
 - Status: preprint on SSRN (No. 7052096).

- **Multi-Agent Framework for Scalable Fleet Maintenance Planning**

- Designed a distributed multi-agent framework that coordinates decisions across large fleets using decentralized MDP formulations.
- Status: dissertation work; manuscript in preparation.

Publications

Google Scholar (October 2026): 46 citations.

Peer-Reviewed Journal Articles

- **A. Salehiyan**, N. Serrano, F. Hernando-Gallego, D. Martín, J. V. Álvarez-Bravo. “A Novel Evolutionary Optimized Transformer-Deep Reinforcement Learning Framework for False Data Injection Detection in Industry 4.0 Smart Water Infrastructures.” *Computers, Materials & Continua*, 87(2), 68, 2026.
- **A. Salehiyan**, P. S. Moghaddam, M. Kaveh. “An Optimized Transformer-GAN-AE for Intrusion Detection in Edge and IIoT Systems: Experimental Insights from WUSTL-IIoT-2021, EdgeIIoTset, and TON_IoT Datasets.” *Future Internet*, 17(7), 279, 2025.

Conference Proceedings

- P. Vaghfi Mohebbi, **A. Salehiyan**, A. Deep. “Disease Cluster Analysis in Electronic Health Records: Insights into Mortality and Comorbidity Patterns.” *Proceedings of the IISE Annual Conference & Expo*, pp. 1–6, 2025.
- “Discovering Hidden Relationships and Patterns in CNC Equipment Failures Using the Apriori Algorithm.” *Civilica*, 2022 (in Persian).

Preprints and Manuscripts Under Review

- **A. Salehiyan**, J. Lee, M. Kim, A. Deep. “Deviation-Informed Maintenance Planning: Cost Implications of Departures from Sensor-Based Optimal Policy.” Submitted to *INFORMS Journal on Data Science*, 2025.
- **A. Salehiyan**, J. Lee, M. Kim, A. Deep. “A Scalable POMDP-based Framework for Real-Time Maintenance Planning using Multiple Sensor Signals.” TechRxiv preprint, 2024.
- **A. Salehiyan**, M. Masae, P. Vichitkunakorn, A. Deep. “A Hierarchical Deep Q-Learning Approach for Picker Routing in Two-Block Warehouses.” SSRN preprint No. 7052096.

In Preparation

- **A. Salehiyan**, A. Deep. “A Multi-Agent Framework for Scalable Fleet Maintenance Planning under System Constraints.”
- **A. Salehiyan**, A. S. Dabas, A. Deep. “Deep Reinforcement Learning Framework for Maintenance Planning in Multi-Component Systems with Stochastic Interdependence.”

Conference Presentations

- “Disease Cluster Analysis in Electronic Health Records: Insights into Mortality and Comorbidity Patterns.” *IISE Annual Conference & Expo*, Atlanta, GA, 2025 (Talk).
- “A Multi-Agent Framework for Scalable Fleet Maintenance Planning under System Constraints.” *INFORMS Annual Meeting*, Atlanta, GA, 2025 (Talk).
- “A Scalable Algorithm for Condition-Based Maintenance with High-Dimensional Sensor Data.” *Reliability and Maintainability Symposium (RAMS)*, Miramar Beach, FL, 2025 (Talk).
- “POMDP-Based Optimal Maintenance Planning Using Multiple Sensor Signals.” *INFORMS Annual Meeting*, Seattle, WA, Oct 2024 (Talk).
- “POMDP-Based Optimal Maintenance Planning Using Multiple Sensor Signals.” *OSU Student Research Symposium*, Stillwater, OK, 2024 (Poster; 3rd Place Award).

Teaching Experience

- **Graduate Teaching Assistant / Instructor, Oklahoma State University** 2025 – Present
 - **Fall 2026:** Teaching assistant, ENGR 1412 Introductory Engineering Computer Programming; weekly Excel assignments, grading, and help sessions for lab sections.
 - **Fall 2025:** Taught “Python Programming” to 60 students; algorithms, data visualization, and engineering problem-solving in Python.
 - **Summer 2025:** Taught “Engineering Economics” to 50 students; cash-flow analysis, NPV, IRR, and capital-allocation decisions.
 - **Spring 2025:** Taught “Python for Data Analysis” to 60 undergraduate and graduate students; data structures, numerical computing (NumPy), and statistical programming (pandas).

Industry Experience

- **Industrial Engineer, Karin Crane Company** Apr 2019 – Oct 2019
 - Coordinated process improvements and documented operational workflows across production teams; supported resource planning, scheduling, and quality control.

Honors & Awards

- 3rd Place, OSU Student Research Symposium (Graduate Poster Category), 2024.
- Ranked 9th in M.Sc. (Systems Optimization) cohort, K. N. Toosi University of Technology, 2021.
- Ranked 4th in B.Sc. (Industrial Engineering) cohort, Islamic Azad University, Qazvin, 2019.

Service & Leadership

- Peer reviewer for academic journals in operations research, maintenance engineering, and industrial systems.
- President, Iranian Student Association (IRSA), Oklahoma State University, 2025 – Present.

Professional Memberships

- INFORMS (since 2024); IISE and IISE Reliability & Maintenance Society (since 2024); SME (Society of Manufacturing Engineers).

Languages

- Persian (native); English (full professional proficiency); Spanish (intermediate).